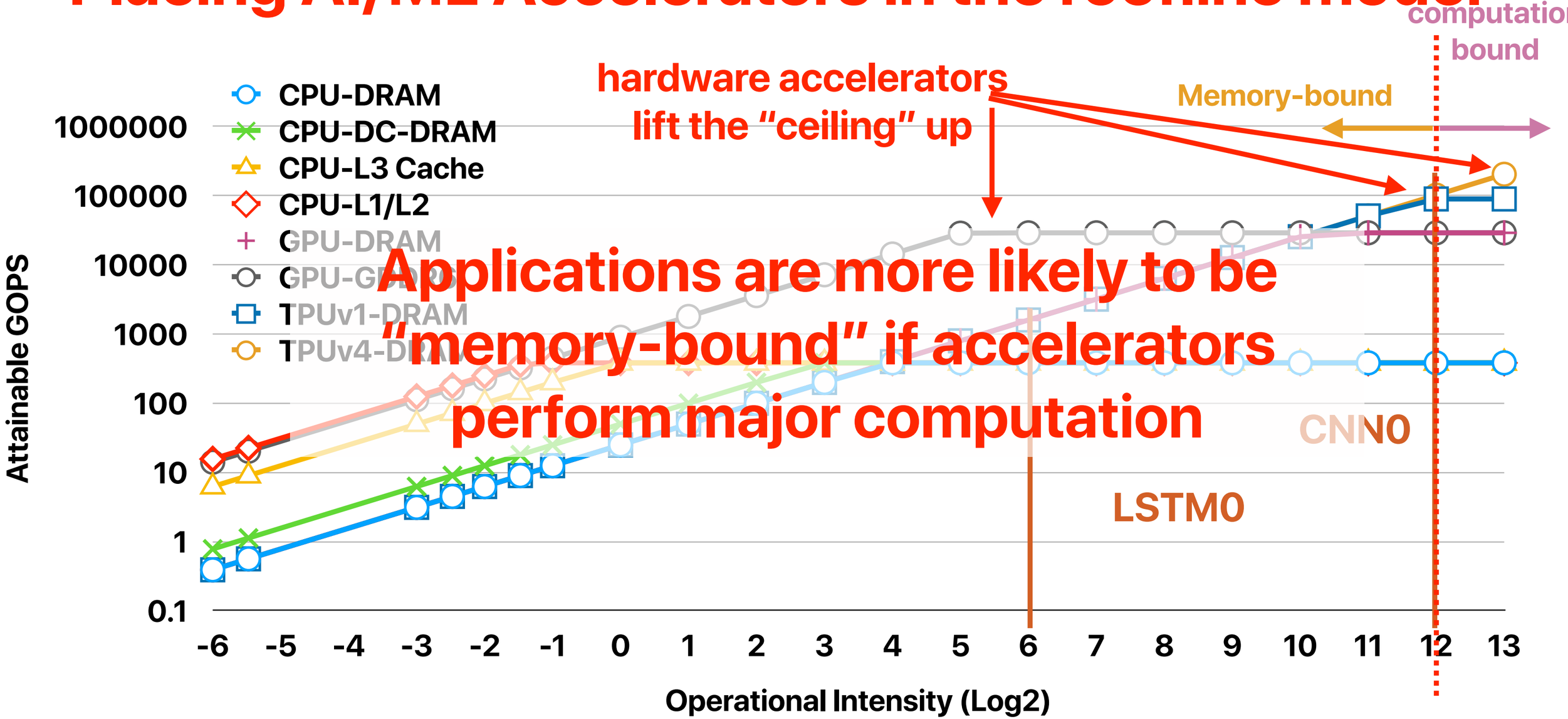
Processing-*-Memory (1)

Hung-Wei Tseng

Placing AI/ML Accelerators in the roofline model



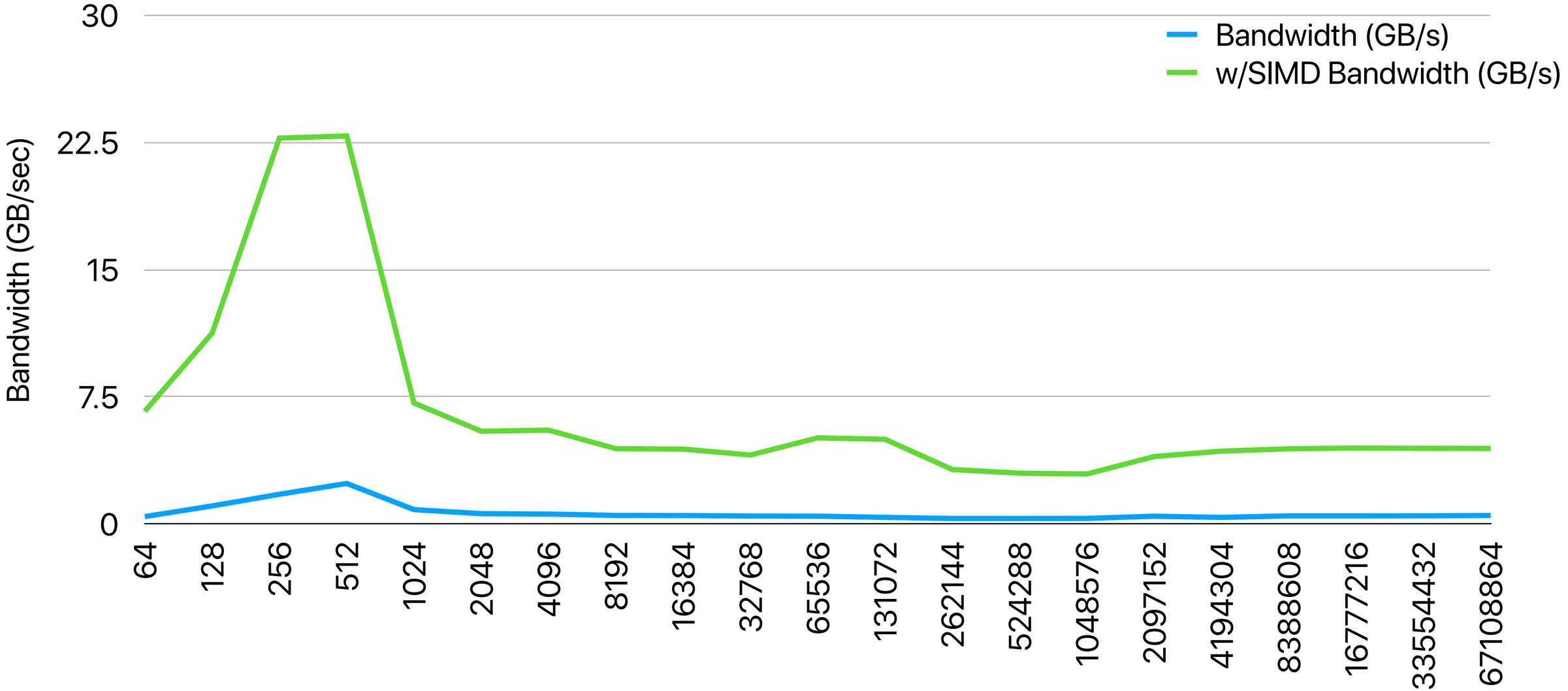
Recap: Memcpy

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/time.h>
#include <time.h> /* for clock_gettime */
#include <malloc.h>

void write_memory_loop(void* array, size_t size) {
  size_t* carray = (size_t*) array;
  size_t i;
  for (i = 0; i < size / sizeof(size_t); i++) {
    carray[i] = 1;
  }
}


int main(int argc, char **argv)
{
    size_t *array;
    size_t *dest;
    size_t size;
    double total_time;
    struct timespec start, end;
    struct timeval time_start, time_end;
    size = atoi(argv[1])/sizeof(size_t);
    array = (size_t *)malloc(sizeof(size_t)*size);
    dest = (size_t *)malloc(sizeof(size_t)*size);
    clock_gettime(CLOCK_MONOTONIC, &start); /* mark start
time */
    #ifdef SIMD
    write_memory_avx(array, size);
    #else
    write_memory_loop(array, size);
    #endif
    clock_gettime(CLOCK_MONOTONIC, &start); /* mark start
time */
    memcpy(dest, array, size*sizeof(size_t));
    clock_gettime(CLOCK_MONOTONIC, &end); /* mark start
time */
    total_time = ((end.tv_sec * 1000000000.0 +
end.tv_nsec) - (start.tv_sec * 1000000000.0 +
start.tv_nsec));
    fprintf(stderr,"Latency: %.0lf ns, GBps: %lf,
%llu\n",total_time, (double)((double)size*sizeof(size_t)/
(total_time)), dest[rand()%size]);
    return 0;
}
```

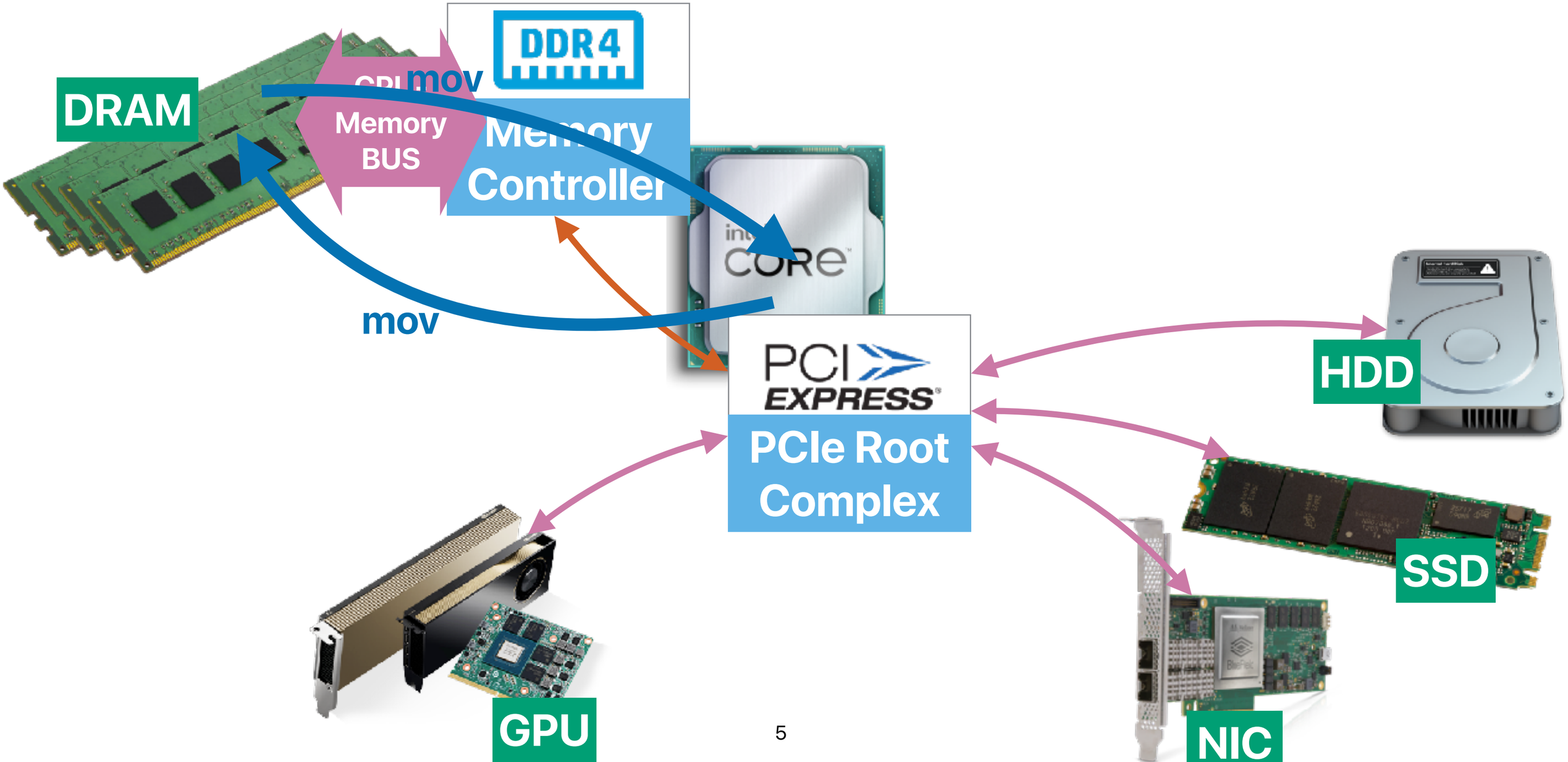
memmove & memcpy: 5% cycles in Google's datacenter

Svilen Kanev, Juan Pablo Darago, Kim Hazelwood,
Parthasarathy Ranganathan, Tipp Moseley, Gu-Yeon Wei, and
David Brooks. ISCA '15

Memory copy bandwidth



Recap: The landscape of modern computers



memcpy in x86

What's the x86 code for memcpy

Regenerate draft

Here's a breakdown of the x86 code for memcpy, incorporating insights from various sources and addressing potential concerns:

Key Considerations:

- **Architecture and Optimization:** The exact code can vary depending on the specific x86 architecture (32-bit vs. 64-bit) and compiler optimizations.
- **Overlapping Memory:** `memcpy` assumes non-overlapping memory regions. For overlapping regions, use `memmove`.

Common Implementation (32-bit x86):

Code snippet

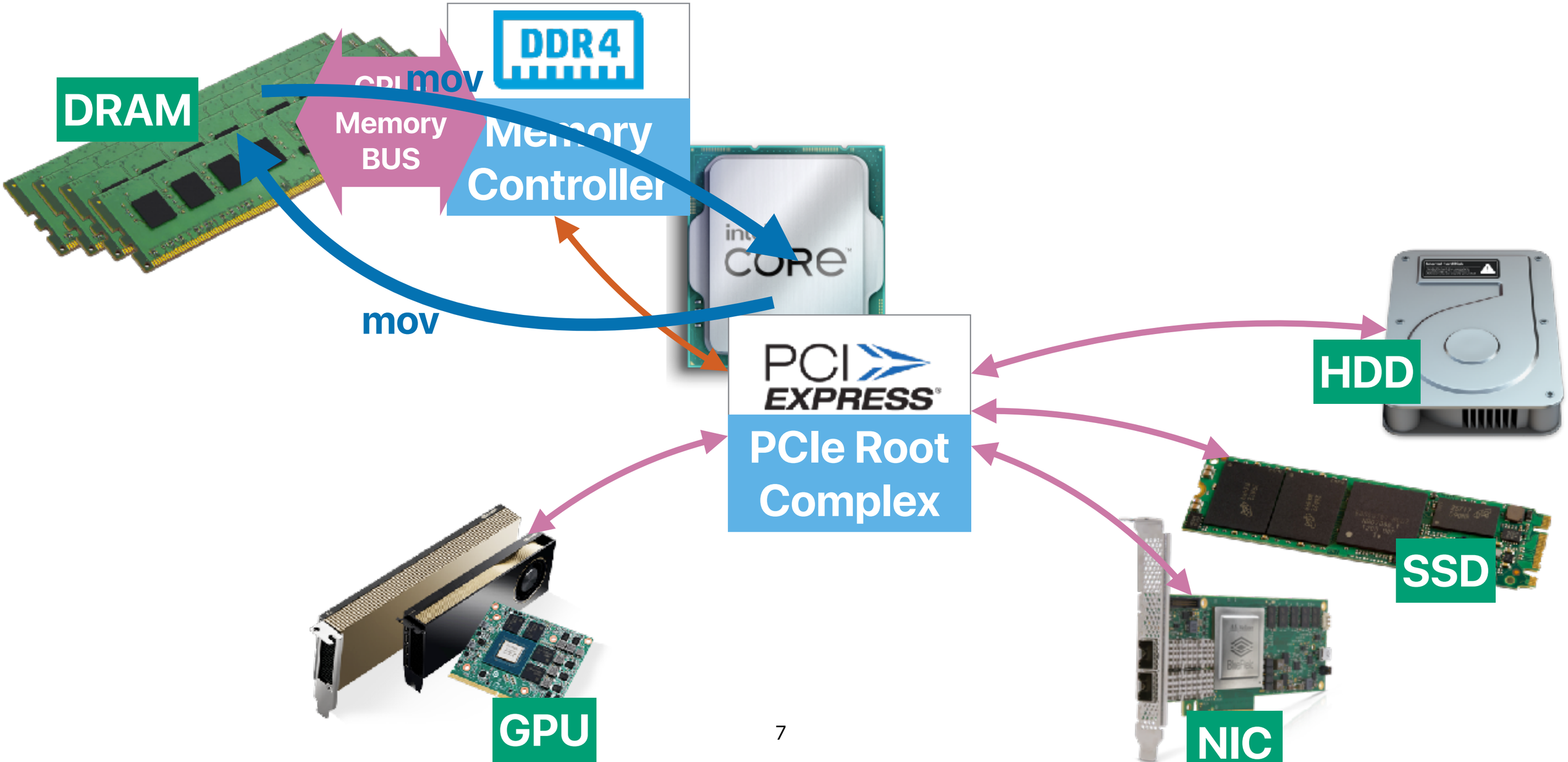
```
memcpy:
    push ebp
    mov ebp, esp

    push esi
    push edi
    push ecx

    mov edi, [ebp + 8]  ; Destination address
    mov esi, [ebp + 12] ; Source address
    mov ecx, [ebp + 16] ; Number of bytes to copy

    rep movsb           ; Copy bytes using movsb instruction

    pop ecx
    pop edi
    pop esi

    pop ebp
    ret
```

Use code with caution. Learn more

Explanation:

1. **Prologue:**
   - Saves registers `ebp`, `esi`, `edi`, and `ecx` on the stack.
2. **Parameter Access:**
   - Loads function arguments (destination, source, and size) into registers.
3. **Main Loop:**
   - `rep movsb`:
     - `rep`: Repeat until `ecx` becomes zero.
     - `movsb`: Move a byte from `esi` to `edi`.
4. **Epilogue:**
   - Restores saved registers.
   - Returns from the function.

Recap: The landscape of modern computers



Why is memcpy slow?

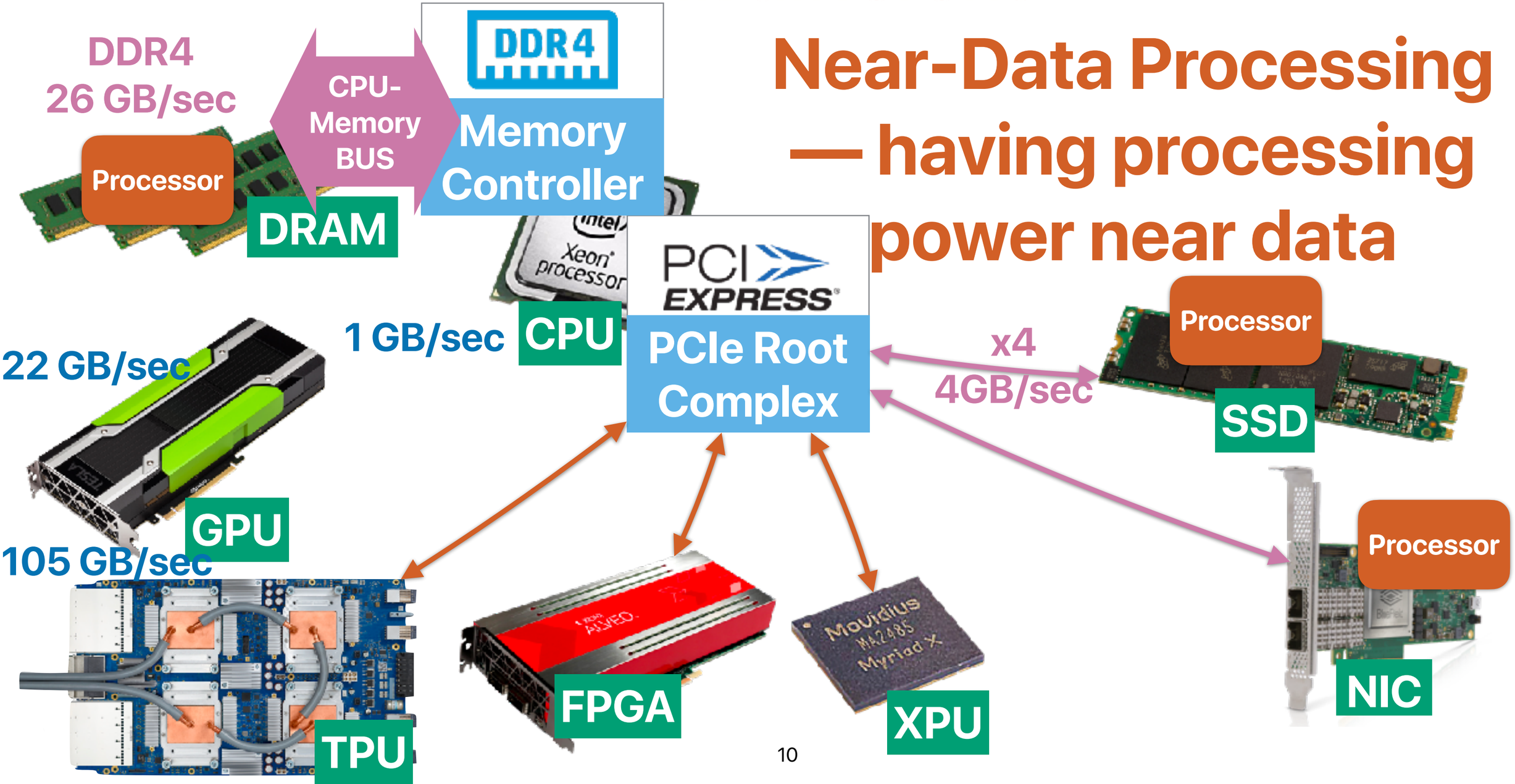
- Each instruction can only move limited amount of data
- Lots of instructions to move data — processor can only handle limited outstanding memory requests
- They must temporarily store data in CPU registers
- Data have to flow through CPU-memory bus twice
- Other overhead
  - Page fault

Outline

- Performance of main memory subsystem (cont.)
- Near/In-memory processing

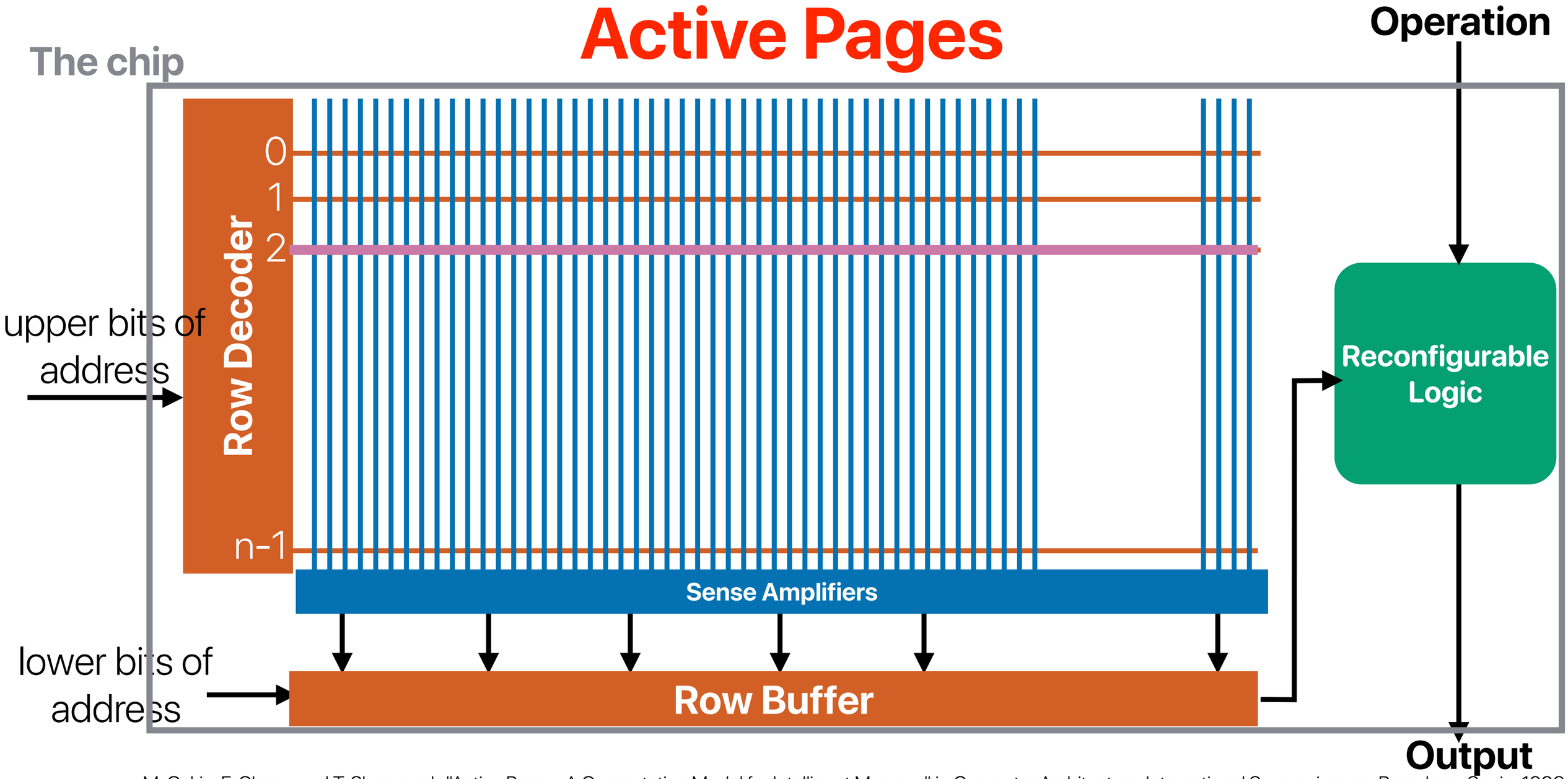
Think different

Near-Data Processing — having processing power near data



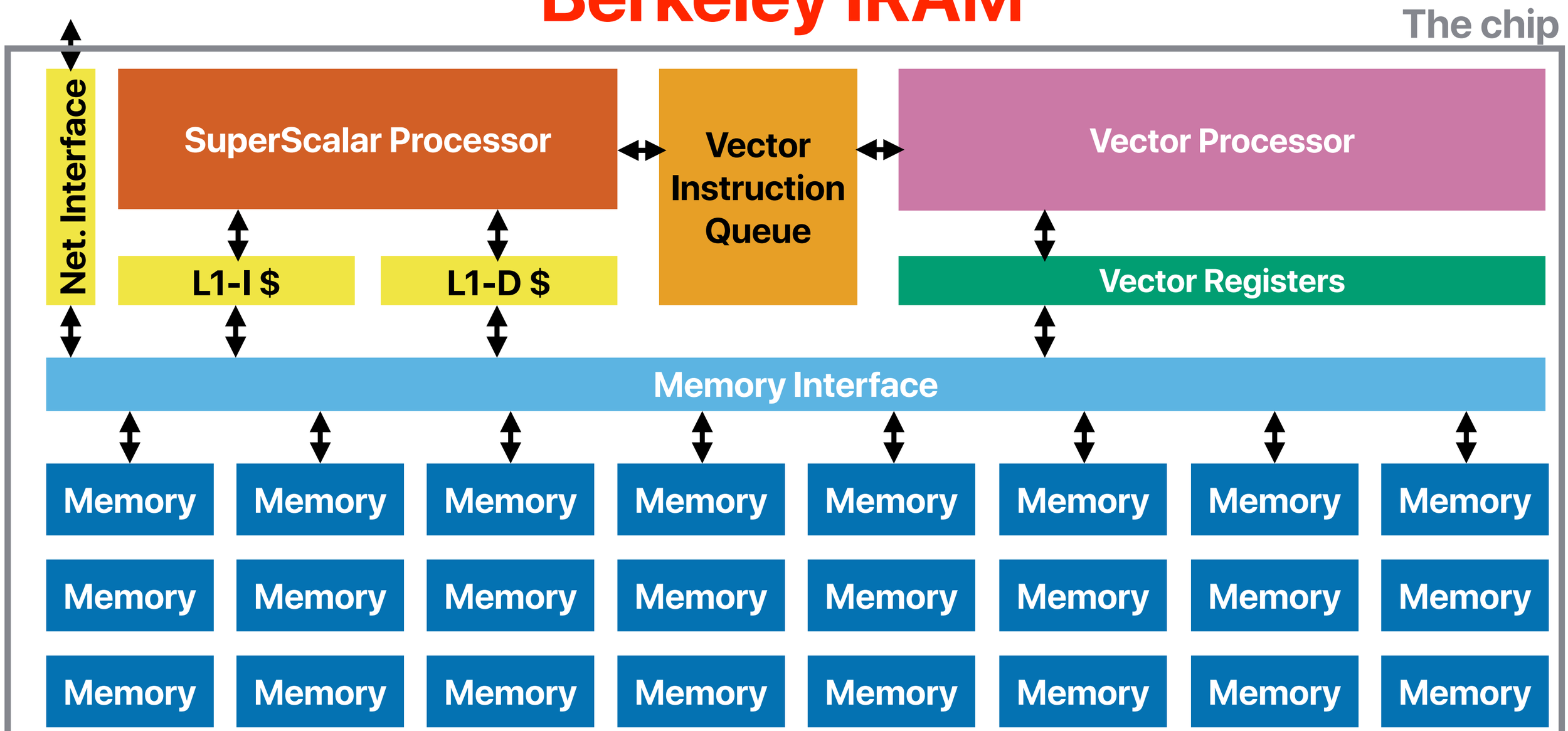
Near-memory processing

Active Pages



M. Oskin, F. Chong and T. Sherwood, "Active Pages: A Computation Model for Intelligent Memory," in Computer Architecture, International Symposium on, Barcelona, Spain, 1998

Berkeley IRAM



David Patterson, Thomas Anderson, Neal Cardwell, Richard Fromm, Kimberly Keeton, Christoforos Kozyrakis, Randi Thomas, and Katherine Yelick. 1997. A Case for Intelligent RAM. IEEE Micro 17, 2 (March 1997), 34–44.

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Architecture-wise, what's the difference between iRAM and ActivePages? What can they improve?

Processing near memory v.s. processing in memory

- Processing near memory (iRAM)
  - Placing processors where data processing can be done without gong through system interconnects
  - Still limited by the bandwidth/latency of the internal device bandwidth
- Processing in memory (ActivePages)
  - Placing processors where data processing can be done without gong through in-device/module interconnects
  - Still limited by the bandwidth/latency of accessing the memory cells
  - May still need multiple accesses to complete an operation

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What kinds of and what applications would fit well in IRAM or Active Pages?

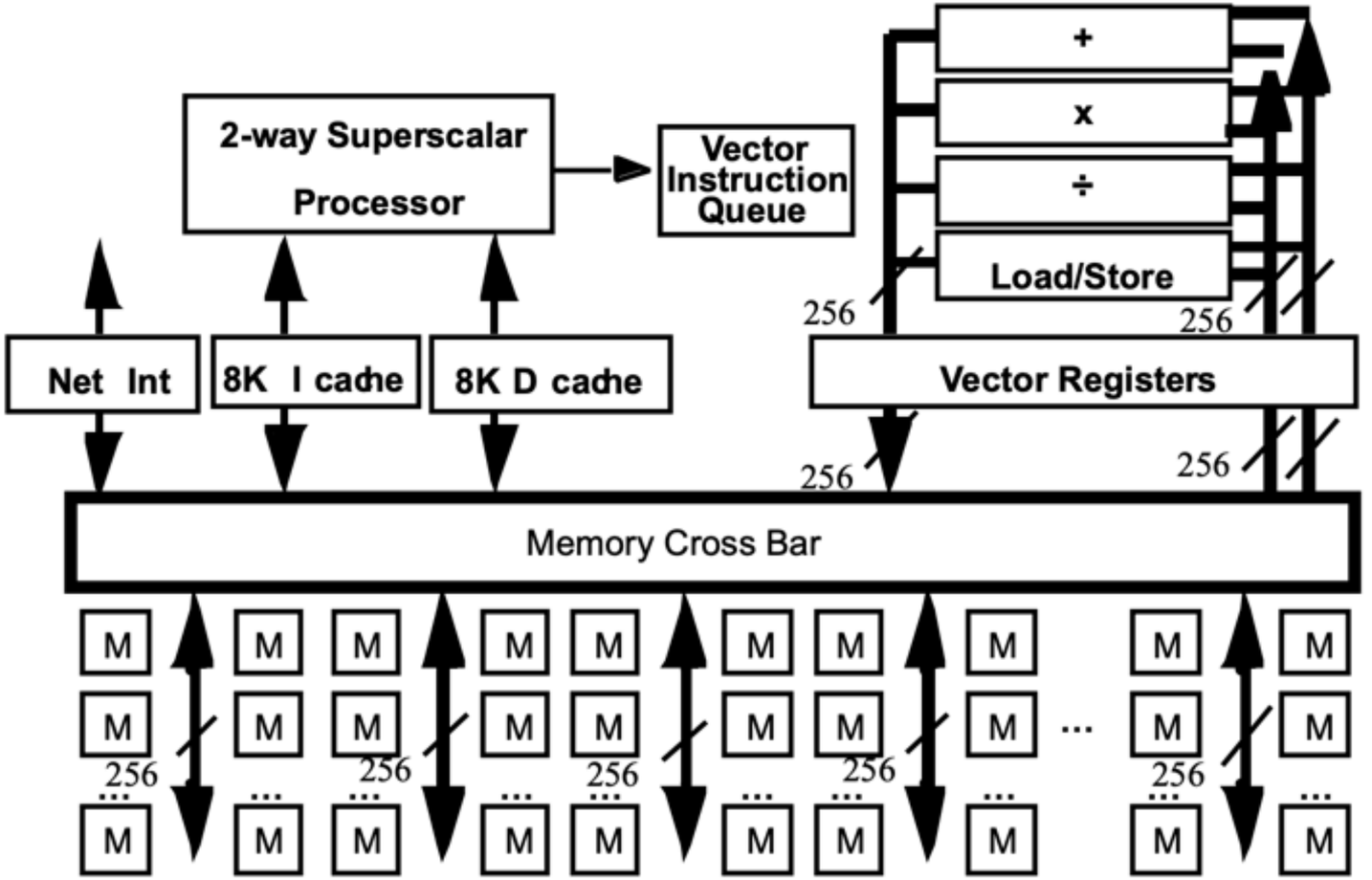
What kind of applications?

- Both Active Pages and IRAM operates on a "page" or "pages" of data simultaneously
- Throughput oriented rather than latency sensitive
- Cannot be cached well
  - DRAM-intensive
  - Low locality (hard-to-predict patterns)
  - Large data footprint

Table 2: CPI, cache misses, and time spent in Alpha 21164 for four programs

Category	SPECint92	SPECfp92	Database	Sparse
Clocks Per Instruction (CPI)	1.2	1.2	3.6	3.0
I cache misses per 1000 instructions	7	2	97	0
D cache misses per 1000 instructions	25	47	82	38
L2 cache misses per 1000 instructions	11	12	119	36
L3 cache misses per 1000 instructions	0	0	13	23
Fraction of time in processor	0.78	0.68	0.23	0.27
Fraction of time in I cache misses	0.03	0.01	0.16	0.00
Fraction of time in D cache misses	0.13	0.23	0.14	0.08
Fraction of time in L2 cache misses	0.05	0.06	0.20	0.07
Fraction of time in L3 cache misses	0.00	0.02	0.27	0.58

0.25 µm, Fast Logic IRAM: ≈500MHz
5 GFLOPS(64b)/40 GOPS(8b)/24MB



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benchmark	Small			Large		
	Conven-	IRAM		Conven-	IRAM	
	tional	(.75 **X**)	(1.0 **X**)	tional	(.75 **X**)	(1.0 **X**)
hsfsys	138	112 *(0.81)*	150 *(1.08)*	149	114 *(0.77)*	152 *(1.02)*
noway	111	99 *(0.89)*	132 *(1.19)*	127	104 *(0.82)*	139 *(1.09)*
nowsort	109	104 *(0.95)*	138 *(1.27)*	136	110 *(0.81)*	147 *(1.08)*
gs	119	107 *(0.90)*	142 *(1.20)*	141	109 *(0.78)*	146 *(1.04)*
ispell	145	113 *(0.78)*	151 *(1.04)*	149	115 *(0.77)*	153 *(1.03)*
compress	91	102 *(1.13)*	137 *(1.50)*	127	104 *(0.82)*	139 *(1.09)*
go	97	96 *(0.99)*	128 *(1.31)*	128	98 *(0.76)*	130 *(1.02)*
perl	136	106 *(0.78)*	141 *(1.04)*	140	107 *(0.76)*	142 *(1.01)*

Table 6: **Performance (in MIPS) of IRAM versus conventional processors, as a function of processor slowdown in a DRAM process.** Only the models with the 32:1 DRAM to SRAM-cache area density ratio are shown. The values in parentheses are the ratios of performances of the IRAM models compared to the CONVENTIONAL implementations. Ratios greater than 1.0 indicate that IRAM has higher performance.

Active Pages: Applications

Memory-Centric Applications			
Name	Application	Processor Computation	Active Page Computation
Array	C++ standard template library array class	C++ code using array class Cross-page moves	Array insert, delete, and find
Database	Address Database	Initiates queries Summarizes results	Searches unindexed data
Median	Median filter for images	Image I/O	Median of neighboring pixels
Dynamic Prog	Protein sequence matching	Backtracking	Compute MINs and fills table
Processor-Centric Applications			
Name	Application	Processor Computation	Active Page Computation
Matrix	Matrix multiply for Simplex and finite element	Floating point multiplies	Index comparison and gather/scatter of data
MPEG-MMX	MPEG decoder using MMX instructions	MMX dispatch Discrete cosine transform	MMX instructions

Table 2: Summary of partitioning of applications between processor and active pages

Active Pages

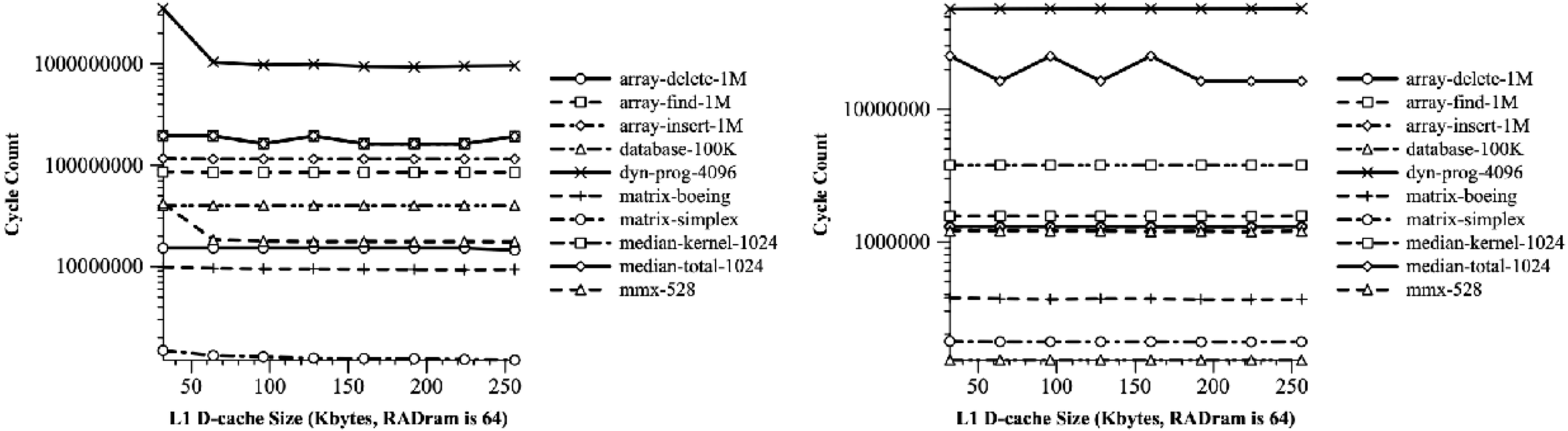

Figure 5: Conventional **(left)** and RADram **(right)** Execution Time vs. L1 Data Cache Size

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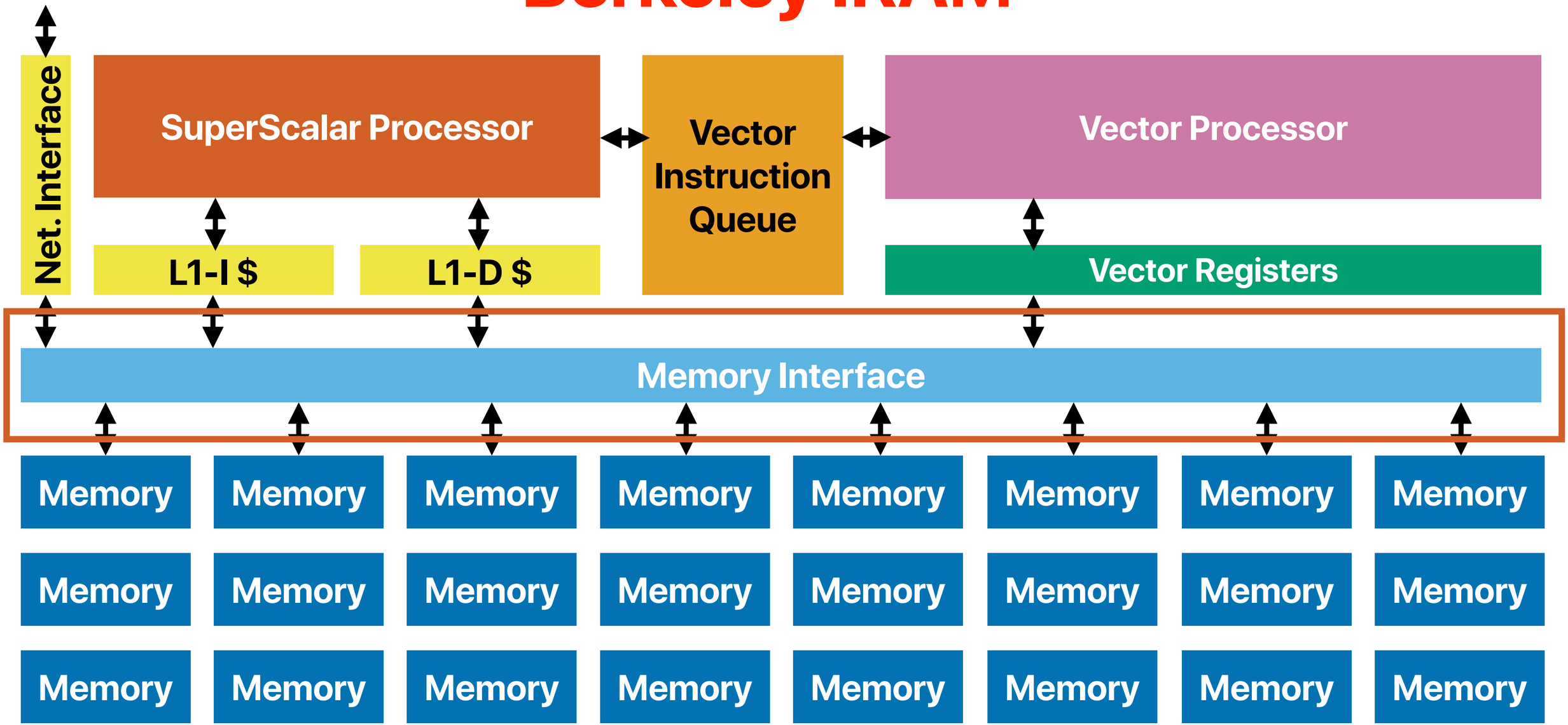
Why “active pages” and “iRAM” did not take away during that era?

Why IRAM or Active Pages does not work out?

- Programming
  - Why GPU succeed later?
- Applications
- Lack of industrial implementation, support
- Hardware design — Processor v.s. DRAM process technologies
- Cost
  - For Active Pages — small processor each module
  - For IRAM
    - High bandwidth on-chip interconnect was expensive
    - Other hardware design issues
      - Significant differences in clock rates among units

Berkeley IRAM



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NVIDIA Grace Hopper GH200

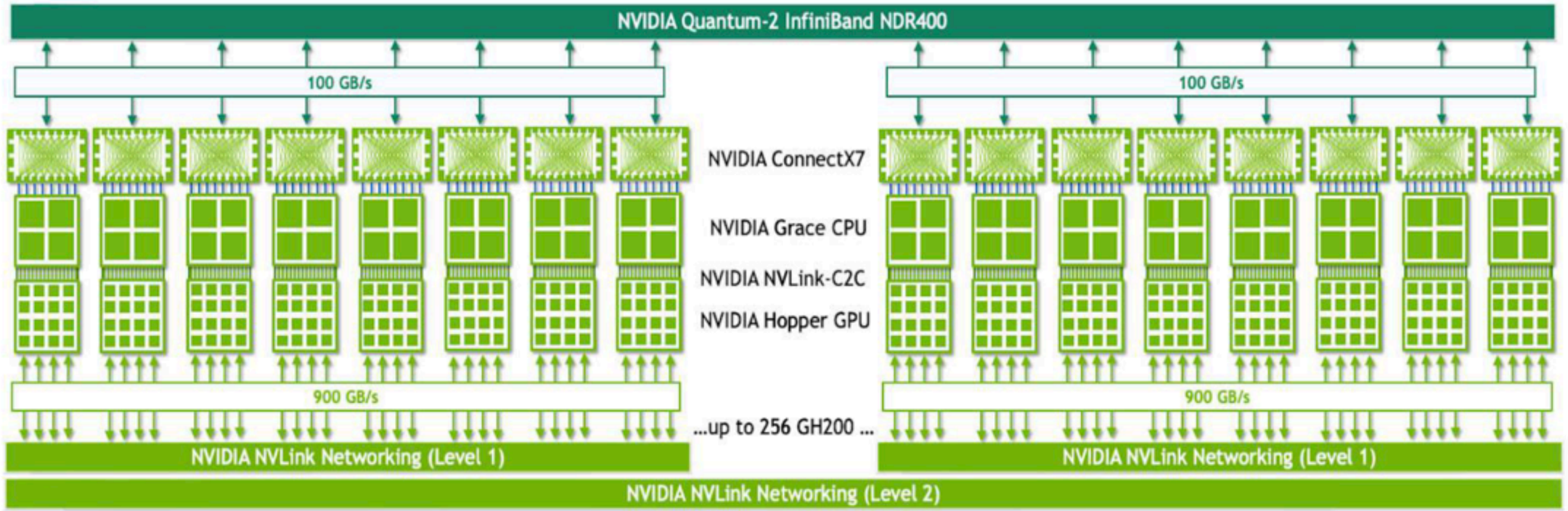


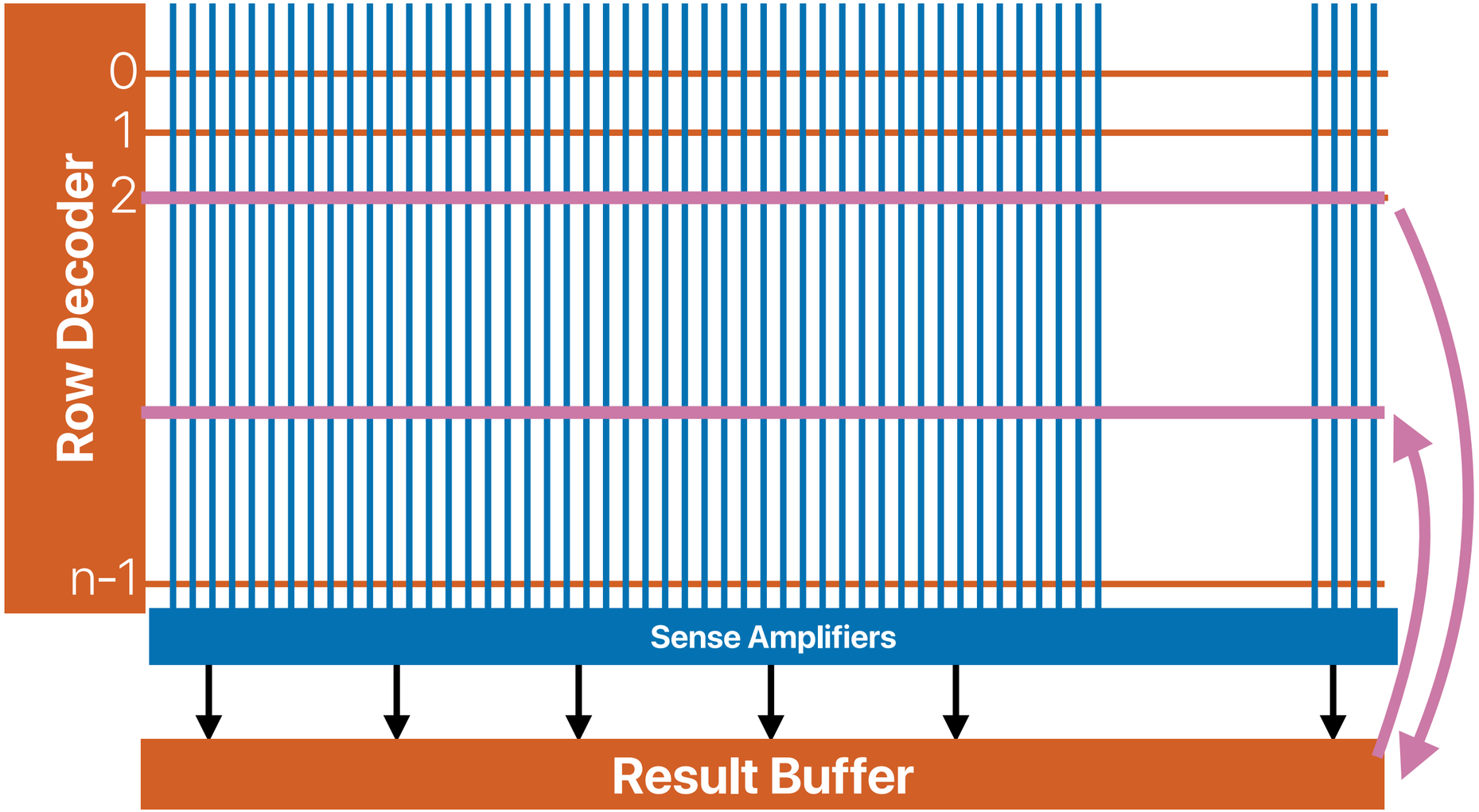
Figure 4. NVIDIA GH200 NVL32 with NVLink Switch System for strong-scaling giant ML workloads

3D-die stacking

- Providing huge bandwidth between memory chips and processors (e.g., HBM)
- The renaissance of near-memory processing!
  - Zhu, Qiuling, Berkin Akin, H. Ekin Sumbul, Fazle Sadi, James C. Hoe, Larry Pileggi, and Franz Franchetti. A 3D-stacked logic-in-memory accelerator for application-specific data intensive computing.In 3DIC. 2013.
  - Dongping Zhang, Nuwan Jayasena, Alexander Lyashevsky, Joseph L. Greathouse, Lifan Xu, and Michael Ignatowski. TOP-PIM: throughput-oriented programmable processing in memory. HPDC '14. 2014
  - Farmahini-Farahani, Amin, Jung Ho Ahn, Katherine Morrow, and Nam Sung Kim. NDA: Near-DRAM acceleration architecture leveraging commodity DRAM devices and standard memory modules. In HPCA. 2015.
  - Junwhan Ahn, Sungpack Hong, Sungjoo Yoo, Onur Mutlu, and Kiyoung Choi. A scalable processing-in-memory accelerator for parallel graph processing. ISCA '15. 2015.
  - Youngeun Kwon, Yunjae Lee, and Minsoo Rhu. TensorDIMM: A Practical Near-Memory Processing Architecture for Embeddings and Tensor Operations in Deep Learning. In MICRO '52. 2019

Specialized Processing-In-Memory

Or we just clone/move?

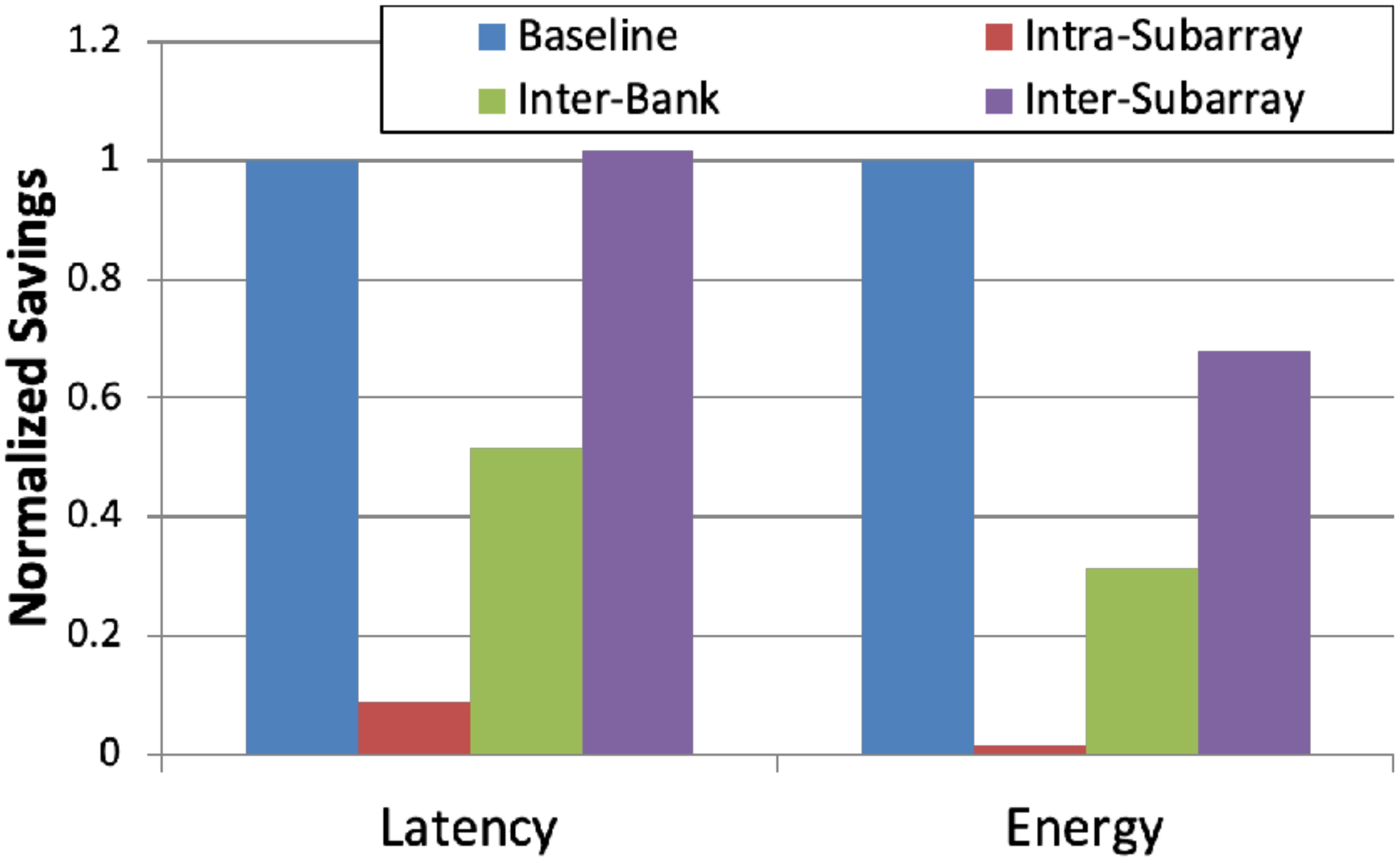


memmove & memcpy: 5% cycles in Google's datacenter

Svilen Kanev, Juan Pablo Darago, Kim Hazelwood, Parthasarathy Ranganathan, Tipp Moseley, Gu-Yeon Wei, and David Brooks.Profiling a warehouse-scale computer ISCA '15

Vivek Seshadri, Yoongu Kim, Chris Fallin, Donghyuk Lee, Rachata Ausavarungnirun, Gennady Pekhimenko, Yixin Luo, Onur Mutlu, Phillip B. Gibbons, Michael A. Kozuch, and Todd C. Mowry. RowClone: fast and energy-efficient in-DRAM bulk data copy and initialization. In MICRO-46.

The effect of RowClone



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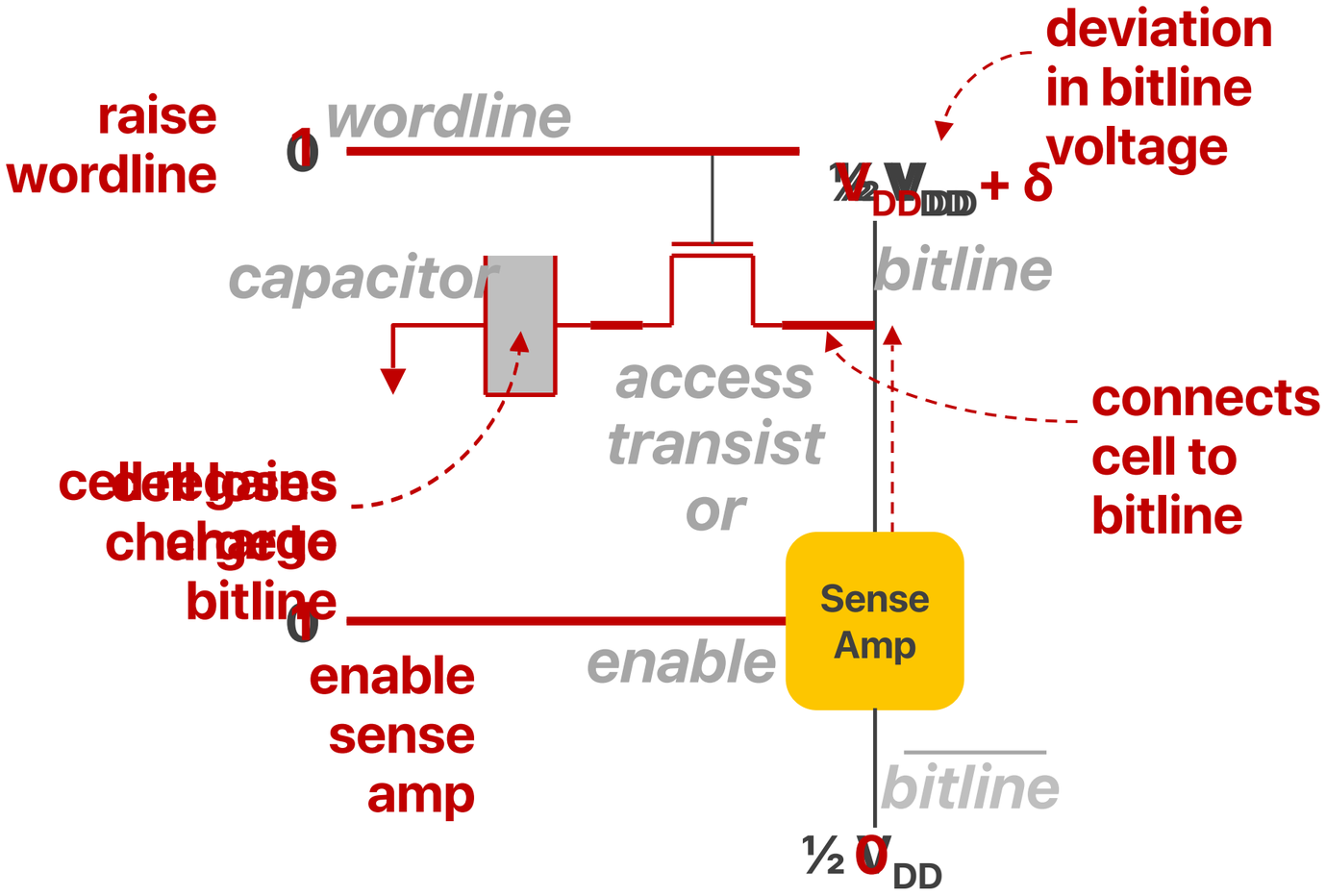
Do you think RowClone is a good idea? Why or why not?

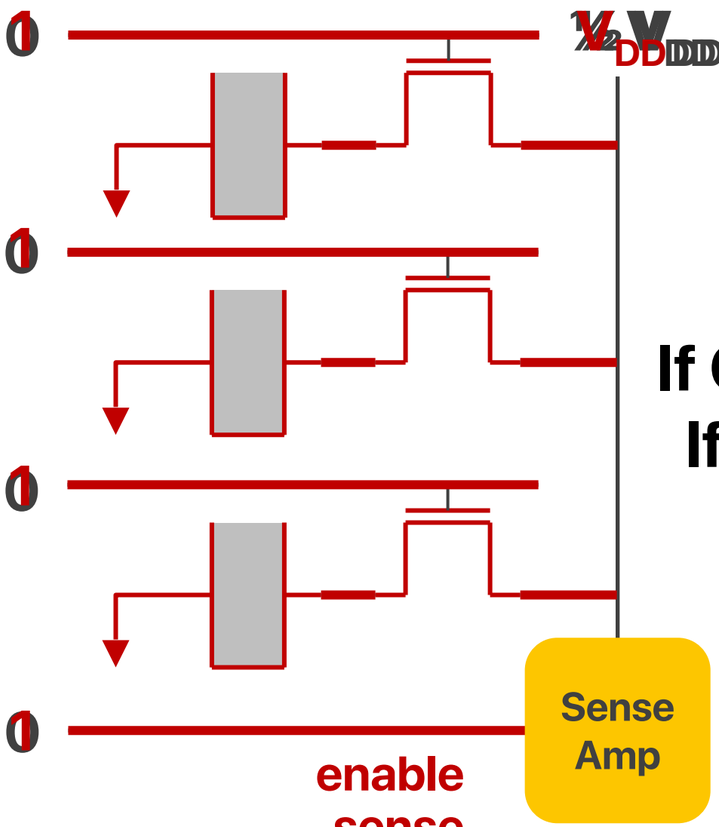
Limitations of RowClone

- Limited functionality — can only perform RowClone
- Data mapping — what if pages are not within the same chip?
- Hardware design — additional cost

Basic DRAM Cell Operation



Activate Three Rows



AB + BC + AC =
C(A+B) + C′AB

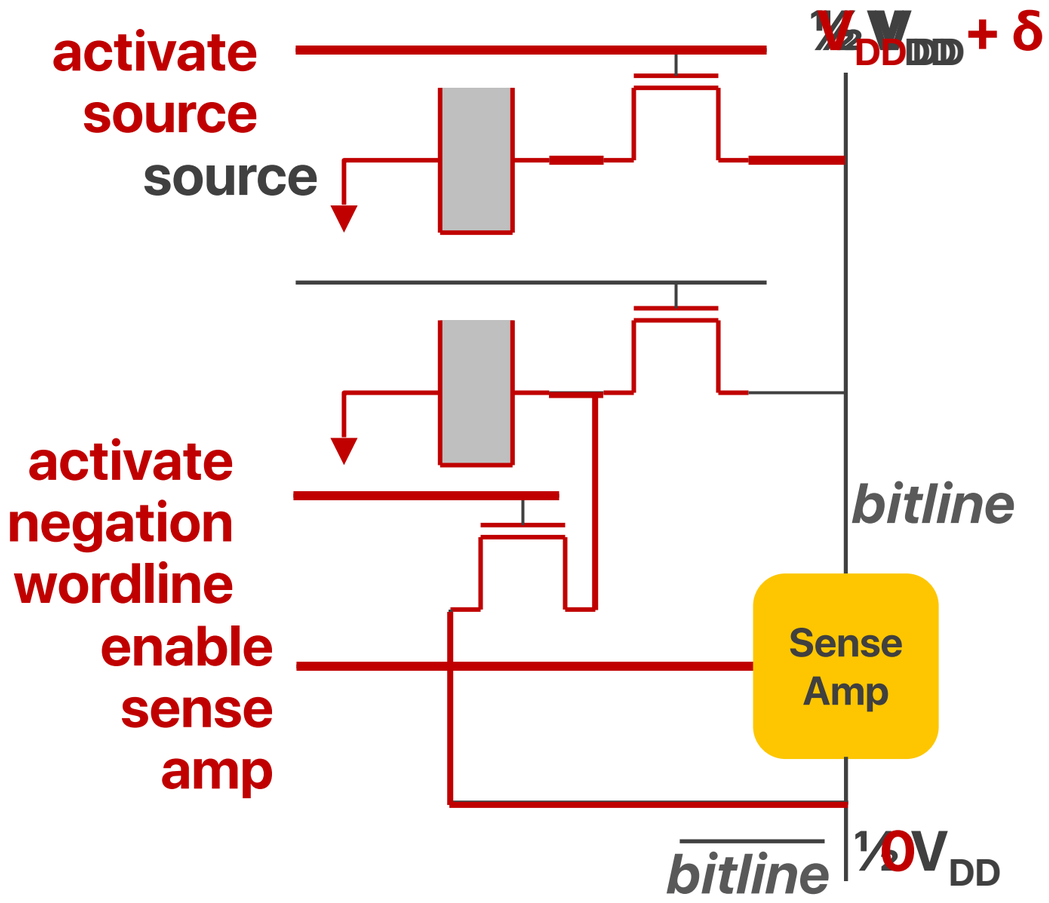
If C is 0, we can compute AND
If C is 1, we can compute OR

Input			Output
A	B	C	
0	0	0	**0**
0	1	0	**0**
1	0	0	**0**
1	1	0	**1**
0	0	1	**0**
0	1	1	**1**
1	0	1	**1**
1	1	1	**1**

		A′B′	A′B	AB	AB′
	Out(A, B)	0,0	0,1	1,1	1,0
C′	0	0	0	1	0
C	1	0	1	1	1

BC AB AC

We can also do NOT



What's the meaning of the ability to perform NAND and NOR?

NAND and NOR are "universal gates" that you can achieve all logical functions with them!

Processing Using Memory

Ambit

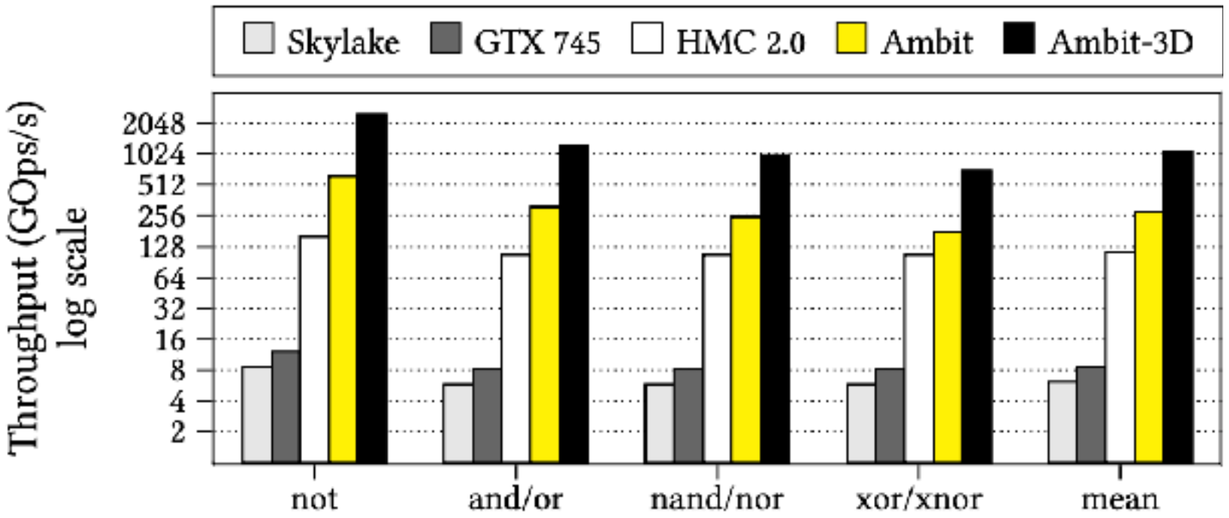


Figure 9: Throughput of bulk bitwise operations.

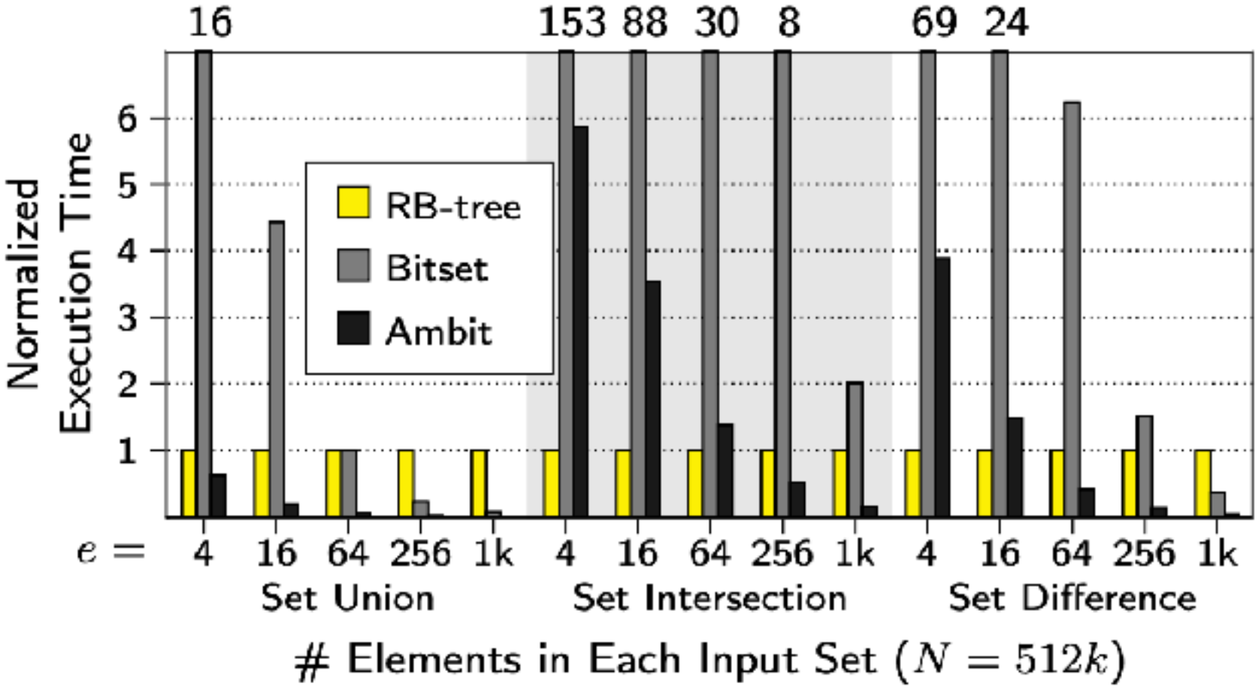


Figure 12: Performance of set operations

Vivek Seshadri, Donghyuk Lee, Thomas Mullins, Hasan Hassan, Amirali Boroumand, Jeremie Kim, Michael A. Kozuch, Onur Mutlu, Phillip B. Gibbons, and Todd C. Mowry. Ambit: in-memory accelerator for bulk bitwise operations using commodity DRAM technology. MICRO-50.

Limitations of processing-using-memory

- Programming
- Hardware design — not that many operations are available
- The memory content will be destroyed after computation
  - The device is no more a “memory” device, but simply an accelerator

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